```
exps]).detach()
            jmeans = torch.stack([ex[2][j] for ex in
exps]).detach()
            vs = torch.stack([ex[3][j] for ex in exps]).detach()
            qvals = torch.stack([

qfunc(jmeans[h].detach(),params[0],layers=layers) \
                                  for h in range(batch_size)])
            target = qvals.clone().detach()
            target[:,torch.argmax(jacts,dim=1)] = jrewards + gamma
* vs
            loss = torch.sum(torch.pow(qvals - target.detach(),2))
            losses[j].append(loss.item())
            loss.backward()
            with torch.no_grad():
                params[0] = params[0] - lr * params[0].grad
            params[0].requires_grad = True
```

- ***1*** num_iter controls how many times we iterate to get rid of the initial randomness from the mean field actions.
- ***2*** Makes a list of lists to store the losses for each agent
- ***3*** replay_size controls the total number of experiences we store in the experience replay list.
- ***4*** The experience replay is a deque collection, which is basically a list with a maximum size.
- ***5*** Sets the batch size to 10, so we get a random subset of 10 experiences from the replay and train with that
- ***6*** The discount factor
- ***7*** Stores the mean field actions for all the agents
- ***8*** Stores the Q values for the next state after taking an action
- ***9*** Since mean fields are initialized randomly, we iterate a few times to dilute the initial randomness.
- ***10*** Iterates through all agents in the grid
- ***11*** Collects an experience and adds to the experience replay buffer
- ***12*** Once the experience replay buffer has more experiences than the batch size parameter, starts training
- ***13*** Generates a list of random indices to subset the replay buffer

That’s a lot of code, but it’s only a little more complicated than what we had for the 1D Ising model. The first thing to point out is that since the mean field of each agent depends on its neighbors, and the neighbors’ spins are randomly initialized, all the mean fields will be random to begin with too.